

CONTACT

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LINKS

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ruben-portfolio-six.vercel.app

SUMMARY

Computer Engineering graduate (July 2026) from the Polytechnic Institute of Bragança, with a peer-reviewed paper on LiDAR sensor modelling accepted at TEEM 2026 (Springer). Builds backend services and LLM-powered applications with Python, FastAPI, Docker, LangChain and RAG. International experience through Erasmus (Croatia) and a BIP programme (Lithuania). Looking for a Software Engineer or AI Engineer role.

LANGUAGES

Portuguese — Native
English — C1 (Advanced)

RUBEN MARCELO BAPTISTA SILVA

EDUCATION

BSc (Licenciatura) in Computer Engineering
Polytechnic Institute of Bragança (ESTIG/IPB) | Sep 2022 – Jul 2026

- Main areas: Programming, Web Development, Distributed Systems, Networks, Cybersecurity. Final-year project defended on 1 July 2026.

FINAL-YEAR PROJECT

Computational Study and Modelling of the LiDAR Sensor on iOS Devices
ESTIG/IPB | 2025 – 2026 | GitHub

- Developed a full iOS 3D-scanning application (Swift, SwiftUI, ARKit) with three capture modes: environment mesh (OBJ), filtered object point cloud (CSV/PLY) and photogrammetry dataset capture.
- Built a Python analysis pipeline (NumPy, SciPy, Matplotlib) and derived an empirical power-law noise model for the Apple LiDAR sensor ($R^2 = 0.97$).
- Photogrammetric reconstruction of captured datasets in Meshroom (CUDA/GPU).

PUBLICATIONS

- Paper on LiDAR sensor characterisation and 3D scanning on iOS — **accepted** at **TEEM 2026** (Porto, Oct. 2026; Springer proceedings).
- Paper from the same research line — submitted to **ARTIIS 2026** (Lisbon, Nov. 2026; under review).

PROJECTS

Document Assistant — RAG Application
Python, LangChain, ChromaDB, Streamlit, Groq | GitHub

- Q&A assistant over user documents: persistent ingestion and vector search (ChromaDB), LLM answers via Groq, Streamlit web UI. In active development.

FastApi-Auth-Rest — Task Manager REST API
Python, FastAPI, MySQL, JWT, Docker | GitHub

- REST API with JWT auth and MySQL, containerised with docker-compose and deployed on Railway; React client in FastApi-Auth-React.

Personal Portfolio Website
Next.js, TypeScript, Tailwind CSS, Vercel | Live site

- Next.js App Router site showcasing my projects. LLM exercises (chains, memory, RAG, agents) in learning-llms.

EXPERIENCE

BIP — Vilnius College, Design & Technical Faculty
Vilnius, Lithuania | May 2025

- Practical robotics project using a robotic arm, integrating software and hardware; handled team communications and troubleshot system faults.

Erasmus Programme — University of Split
Split, Croatia | Feb 2024 – Jul 2024

- One-semester international exchange, improving autonomy, adaptability and teamwork.

TECHNICAL SKILLS

- **Backend:** Python (FastAPI), Java, REST APIs, JWT authentication, MySQL
- **AI / LLMs:** LangChain, RAG, ChromaDB, agents, Hugging Face, Groq
- **Web:** Next.js, React, TypeScript, Tailwind CSS, HTML, CSS, JavaScript
- **Mobile / AR:** Swift, SwiftUI, ARKit
- **Other languages:** C/C++, C#; Python data stack (NumPy, SciPy, Matplotlib)
- **Tools:** Git/GitLab, Docker (Compose), Linux, Railway, Vercel, VMware Workstation, Xcode, LaTeX